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Sweat and air permeable electronics enabled by engineered hierarchical fabric system for exercise management

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Abstract

Scientific exercise monitoring is significant for injury risk prevention and training outcome promotion. Wearable biosensing technologies have emerged as transformative tools for real-time, in-situ physiological state profiling through dynamic biomarker detection during exercise activities. However, current studies remain suboptimal for practical exercise management due to inherent constraints including single-analyte detection paradigms, limited permeability and breathability, and inadequate thermoregulatory performance. Here, we present a novel wearable composite fabric system engineered for multiplex sweat biomarker monitoring while delivering unprecedented wear comfort. The hierarchical architecture was achieved through strategic integration of interwoven fiber-based sensor arrays with conventional textiles, augmented by bilateral deposition of microbead-enhanced polyvinylidene difluoride (PVDF) and polyacrylonitrile (PAN) electrospun nanofiber membranes. The porous matrix, Janus hierarchical gradient, and microbead-mediated interfacial engineering render this fabric system with excellent breathability of 13 mm/s, water vapor transmission rate of 468.9 g/m²/h, and solar reflectance of 96.2%. Systematic validation revealed the system's capabilities in multiplex biomarker tracking during diverse exercise scenarios, machine learning-powered fatigue assessment, and scientific exercise regimen evaluation. This work establishes a universal framework for developing wearable platforms with reliable sensing functionality and wear comfort, facilitating effective personalized exercise healthcare management.

Introduction

Physical exercise serves as a cornerstone of preventive healthcare, delivering benefits spanning physical robustness to mental well-being resilience^{1,2}. However, inappropriate exercise activities often lead to injuries or compromised training outcomes, underscoring the need for personalized exercise health state monitoring and

analysis³. Conventional approaches relying on post-hoc clinical evaluations or retrospective video analysis fundamentally lack the temporal resolution required for dynamic physiological state assessment during exercise, thereby precluding timely intervention and data-driven training optimization.

Physiological biomarkers including physical indicators and chemical analytes in secretions can provide critical insights for exercise health state dynamics^{4,5}. For instance, cardiac function can be quantified through heart rate (HR) monitoring⁶, while electrolyte balance (Na⁺/K⁺ concentrations) and lactic acid (LA) levels in sweat offer vital information regarding hydration status and muscular fatigue^{7–11}, respectively. The pH value correlates with metabolic conditions while uric acid (UA) can serve as an important marker of oxidative stress and suggest the risk

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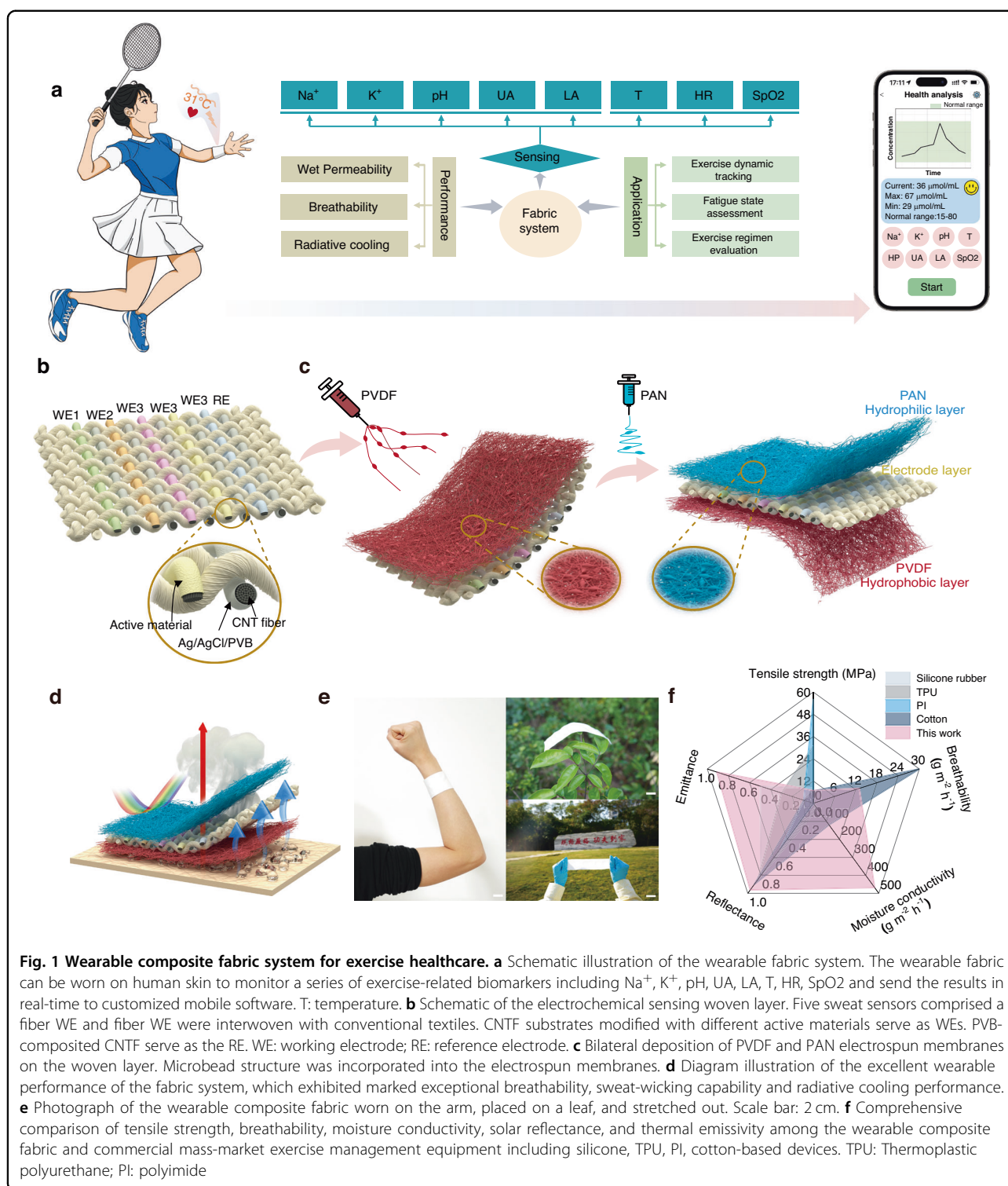
of hyperuricemia and gout^{12,13}. However, conventional approaches employing benchtop instruments like biochemical analyzers and chromatography systems suffer from critical limitations including operational complexity, temporal latency, and lack of portability, rendering them unsuitable for real-time exercise monitoring. This gap underscores the pressing need for effective biosensing platforms capable of continuous biomarker tracking during physical activities.

Recent advancements in wearable biosensor technology have demonstrated remarkable potential for on-body physiological monitoring^{14,15}. While commercial exercise sensors predominantly measure basic physical indicators through rigid electronic components, their mechanical mismatch with human tissue compromises both signal stability and long-term wearability. The emergence of flexible electronics has enabled development of soft sensors with improved conformality^{16–19}. For instance, Cai et al. designed a wearable sensing platform by utilizing N-doped porous carbon and reduced graphene oxide to build the solid contact interface²⁰. The sensing platform achieved ultralow interface diffusion potential and high interface capacitance, enabling sensitive detection of Na^+ , K^+ , and Ca^{2+} in sweat. An integrated smartwatch was further developed, facilitating in situ and real time sweat ions monitoring for exercise healthcare. Gao et al. proposed a fully integrated sensing system by fabricating sensor array on flexible polyethylene terephthalate (PET) substrate and merging commercially available signal driven and transmission technology on a flexible circuit chip¹⁸. The sensor array enabled real time detection of sweat Na^+ , K^+ , glucose, and lactate, and a piezoresistive temperature sensing module was incorporated to realize self-calibration of enzymatic sensors. This work lays the foundation for the development of flexible wearable sweat sensing systems, advancing the development of intelligent exercise healthcare products. Moreover, the exploration of versatile, flexible materials has significantly propelled the development of high-performance wearable sensors. For instance, Wang et al. developed a mesoporous silica@polydopamine-Au based hydrogel for monitoring human motion and sweat ascorbic acid²¹. This nanocomposite hydrogel achieved excellent stretchability, self-healing capability, and mechanical strength. Similarly, Sun et al. reported a bioinspired smart wettable textile system composed of patterned microfoam and nanofiber layer laminate²². Electrochemical sensing array was also incorporated to monitor sweat Na^+ , K^+ , and pH. The whole system possesses brilliant deformability and conformal body integration, which contributed the flexible nanomaterial and structure design.

Despite the diversified advancements in flexible wearable biosensors, largely driven by the rapid progress in

materials science and electronic engineering, critical challenges remain in translating these biosensors into high-performance applications for exercise health management. First, the majority of works focused on monitoring a single or a few exercise-related biomarkers (e.g., ions). These simple analytical architectures fail to capture systemic physiological status, while inadequate sensitivity and selectivity also limit trace biomarker quantification in low-volume secretions. Moreover, exercises are often accompanied by the simultaneous generation of substantial heat and sweat. The accumulation of these byproducts can, on one hand, affect the output signals of sensors and, on the other hand, cause significant discomfort to the user. Simultaneously, maintaining efficient gas exchange between the skin and the external environment is crucial for preserving skin health during exercise. These combined requirements impose stringent demands on wearable sensors intended for exercise health monitoring, necessitating high levels of breathability, permeability, and thermal conductivity. Such performance attributes become particularly critical in scenarios involving extreme heat generation or profuse sweating, such as marathons, mountaineering, or stair climbing. However, conventional wearable products and most existing sensor devices rarely achieve a holistic consideration of these factors, thereby limiting their practical utility in enabling effective and reliable motion health management.

Here, we present an integrated wearable fabric composite system integrating multiplex biosensing capabilities with engineered wear comfort. This system synergistically combines a fiber-based electrochemical sensor array for simultaneous Na^+ , K^+ , pH, LA and UA quantification with a modular accessory physical parameter detection unit monitoring body temperature, HR, percutaneous arterial oxygen saturation (SpO_2). The fabric system demonstrated great capabilities in diverse application scenarios including exercise dynamic tracking, fatigue state assessment, and exercise regimen evaluation, while the system is fully wireless controlled and the results can be real-time transferred to user interface by microcontroller and software engineering (Fig. 1a). Specifically, the fabric system is fabricated by interweaving fiber sensor array with conventional textiles (Fig. 1b), followed by bilateral deposition of polyvinylidene difluoride (PVDF) and polyacrylonitrile (PAN) electrospun nanofiber membranes (Fig. 1c). The sensing module was developed by modifying carbon nanotube fibers (CNTFs) with different functional layers to form two-electrode sensing arrays. Microbead-mediated structures were incorporated in the electrospun membranes to modulate the hydrophilicity/hydrophobicity properties and interfacial characteristics. As a result, the fabric system exhibited excellent breathability



attributed to its porous matrix, and the enhanced Janus hierarchy gradient and altered interfacial architecture contribute to the system's exceptional permeability and radiative cooling performance (Fig. 1d). This flexible, lightweight fabric system enables multiplex analytical

profiling for exercise healthcare, with comprehensively excellent wear comfort for multidimensional exercise management in a prolonged manner (Fig. 1e), while presenting superior wearable performance over conventional wearable exercise sensing platforms (Fig. 1f).

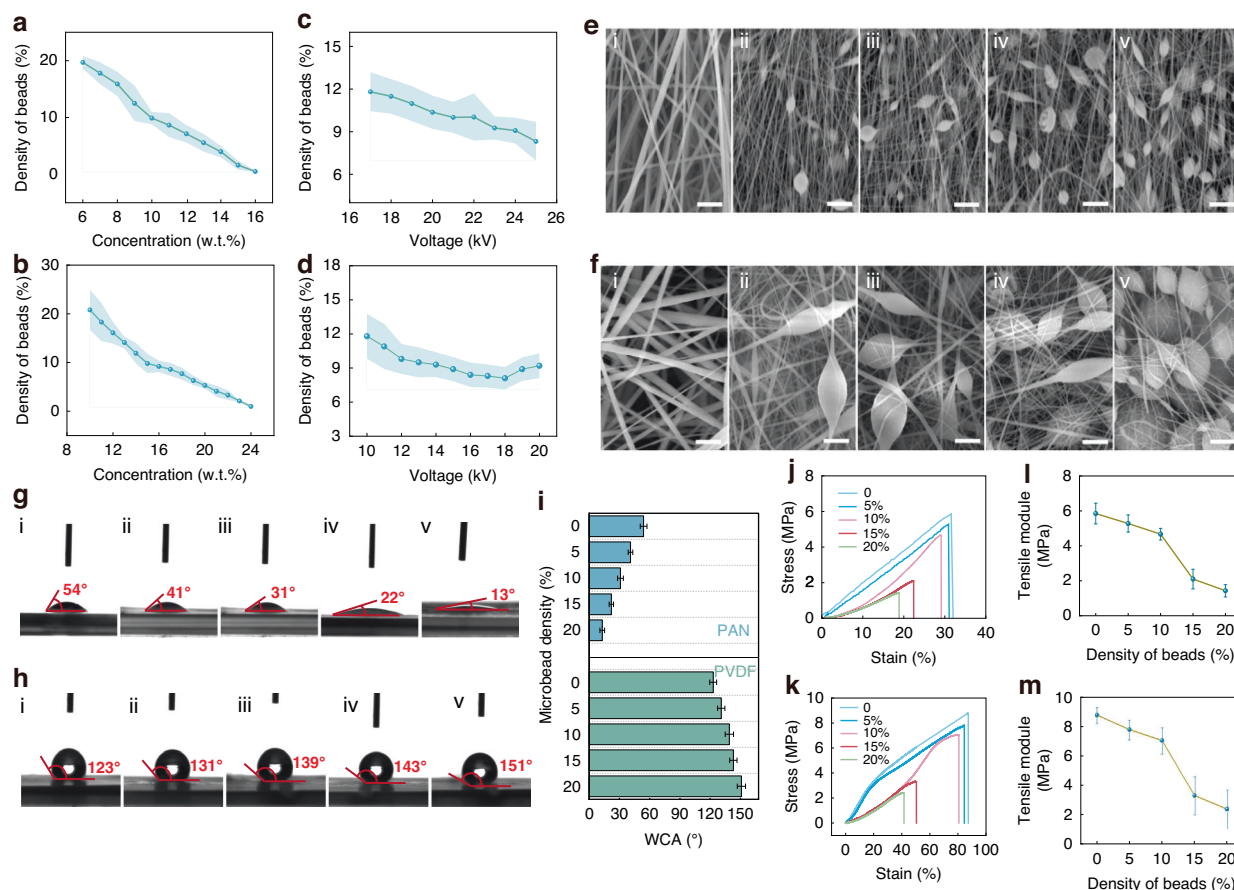


Fig. 2 Characterization of the microbead-mediated electrospun membranes. Modulation of microbead density by polymer concentration for PAN (a) and PVDF (b) membranes. The applied voltages were fixed at 21 kV and 12 kV for electrospinning of PAN and PVDF membranes, respectively. Modulation of microbead density by applied voltage for PAN (c) and PVDF (d) membranes. The polymer concentrations were fixed at 10 wt%, and 15 wt% for electrospinning of PAN and PVDF membranes, respectively. SEM images of PAN (e) and PVDF (f) membranes with (i) 0 wt%, (ii) 5 wt%, (iii) 10 wt%, (iv) 15 wt%, and (v) 20 wt% microbeads. Scale bar: 5 μ m. WCA characterizations of PAN (g) and PVDF (h) membranes with (i) 0 wt%, (ii) 5 wt%, (iii) 10 wt%, (iv) 15 wt%, and (v) 20 wt% microbeads. i Microbead density (%) and WCA (°) of PAN and PVDF membranes with different density of microbeads. The tensile curves of PAN (j) and PVDF (k) membranes with different densities of microbead, and the corresponding tensile strength for PAN (l) and PVDF (m)

Results and discussion

Characterization of microbeads-incorporated electrospun membranes

To fabricate the Janus composite system, the PAN and PVDF layer were deposited on both sides of the porous woven layer by electrospinning techniques, respectively. In the electrospinning process, polymer solutions with adequate chain entanglement capability are formed through solvent dissolution. Under the influence of a high-voltage electrostatic field, these solutions undergo sequential stages including Taylor cone formation, jet motion, solvent evaporation, and solidification, ultimately being collected as fibrous membranes^{23,24}. The extensive stretching of the jet typically results in the formation of continuous and dense fibrous structures. However, Rayleigh instability manifests when alterations occur in molecular entanglement forces or external stresses,

leading to the alternate deposition of microbeads and fibers on the membrane^{24,25}. Our investigations revealed that polymer solution properties predominantly govern this phenomenon. For PAN solutions, bead-free fibrous membranes were obtained at approximately 16 wt% concentration. Progressive reduction in solution concentration correlates with increasing microbead density, reaching approximately 20% at 6 wt% concentration (Fig. 2a). PVDF demonstrated analogous regulatory trends, albeit with reduced concentration sensitivity compared to PAN (Fig. 2b). Bead-free PVDF membranes were achieved at 24 wt% concentration, while microbead density escalated to ~20% at 10 wt%. Notably, microbead density exhibited enhanced concentration sensitivity at the 10 wt% threshold for PVDF. This concentration-dependent behavior originated from diminished molecular entanglement in dilute solutions, which compromises resistance to external

stretching forces, thereby promoting jet fragmentation and subsequent microbead formation through surface tension-induced local contraction^{24,26}.

Voltage intensity also exhibited a moderate influence on microbead formation through modulation of Rayleigh instability. For PAN systems, microbead density decreased approximately at a rate of 0.43%/kV with voltage reduction, attributable to enhanced electric field stabilization of jet motion (Fig. 2c). PVDF demonstrated distinct voltage dependence: microbead density decreased from 11.8% to 8.1% as voltage increased from 10 kV to 18 kV, followed by marginal density elevation at higher voltages (Fig. 2d). This non-monotonic behavior potentially stems from either insufficient entanglement strength to counteract jet splitting at elevated voltages, or amplified jet instability induced by excessive surface charge accumulation^{24,27,28}.

It should be noted that other parameters such as ambient temperature or tip-to-collector distance (TCD) between the spinneret and the collector also possess certain influence on microbead formation. However, compared to polymer concentration and applied voltage, the benefits derived from these parameters are minimal. Flow rate is likewise a critical parameter in the regulation of jet dynamics. But its influence is fundamentally ascribed to affecting the surface charge of the jet, which is analogous to the effects of voltage modulation²⁴. Consequently, these parameters are maintained at constant values during the experiments.

The electrospun membranes with different microbead densities can be obtained for PAN and PVDF (Fig. 2e, f). Results revealed that as the microbead density increased, the morphology of the microbeads within the fiber membrane also underwent changes. With the modulation of spinning parameters affecting Rayleigh instability, elongated protrusions in the localized position of fibers formed microbeads in the electrospun membrane. When the microbead density reached approximately 10%, the microbead shape transitions to an oblate form while remaining consistently connected to the fibers. As the bead density continued to rise, the bead morphology shifted towards a spherical shape and began to detach from the fibers. At around 20% bead density, the spherical beads are nearly completely separated from the fibers, adversely affecting the continuity of the fibers within the membrane, resulting in a more fragmented structure.

Comparative analysis of the hydrophilic/hydrophobic properties in electrospun membranes with varying microbead densities revealed significant structural-performance correlations. The pristine PAN membrane demonstrated moderate hydrophilicity with a water contact angle (WCA) of 54°. Notably, the WCA progressively decreased to approximately 13° at 20% microbead density, which can be attributed to the enhanced actual water contact area and elevated surface tension induced by the

microbead structures^{28,29} (Fig. 2g). In contrast, PVDF membranes exhibited remarkable hydrophobicity, showing a substantial WCA increase from 123° to 151° as microbead density escalated from 0% to 20% (Fig. 2h). These enhanced interfacial properties (Fig. 2i) are beneficial to intensifying the hydrophilic/hydrophobic gradient of Janus hierarchical architecture, facilitating the promotion of moisture-wicking performance. Furthermore, this structural modification-derived contact angle alterations demonstrated superior temporal stability compared to conventional surface treatment methods (Fig. S1). Specifically, plasma-treated samples suffered from rapid hydrophilicity deterioration with significant WCA recovery within only 2 h, while hydrophobic particle-sprayed surfaces regained their original wettability characteristics within approximately 48 h. This comparative study highlights the advantages of intrinsic structural modulation over external surface modifications for achieving durable interfacial control in fibrous membranes.

Subsequent mechanical characterization of both membranes under varying microbead densities was investigated. Tensile testing demonstrated a progressive deterioration in both tensile strength and elongation at break with increasing microbead density, with accelerated degradation observed beyond 10% bead content (Fig. 2j, k). Specifically, PAN membranes exhibited an approximately 20% reduction in tensile modulus when bead density increased from 0% to 10%, followed by a dramatic 55% modulus decrease upon further increment to 15% bead density (Fig. 2l). PVDF membranes displayed analogous deterioration patterns, confirming the universal impact of microbead density on mechanical integrity (Fig. 2m). This phenomenon can be attributed to two primary factors: (1) progressive fiber diameter reduction in high bead-density regions, and (2) compromised fiber continuity beyond 10% microbead density, where isolated microbeads and discontinuous fibers collectively weaken stress distribution efficiency^{23,24}. Given the paramount importance of mechanical robustness for wearable device applications—particularly in motion-intensive health monitoring scenarios—we strategically selected 10% bead density membranes for subsequent device fabrication (This microbead-mediated Janus is defined as MMJ). This optimization balances hydrophilicity/hydrophobicity regulation with sufficient mechanical resilience, addressing both water transmission requirements and operational durability demands in practical implementations.

Characterization of the wearable performance of the MMJ

The Janus hybrid structure was formed due to the contrast hydrophilic/phobic properties of the PVDF and PAN membranes, which can effectively facilitate sweat transportation^{30,31}. As illustrated in Fig. 3a, the secreted sweat first contacts the hydrophobic PVDF layer adjacent

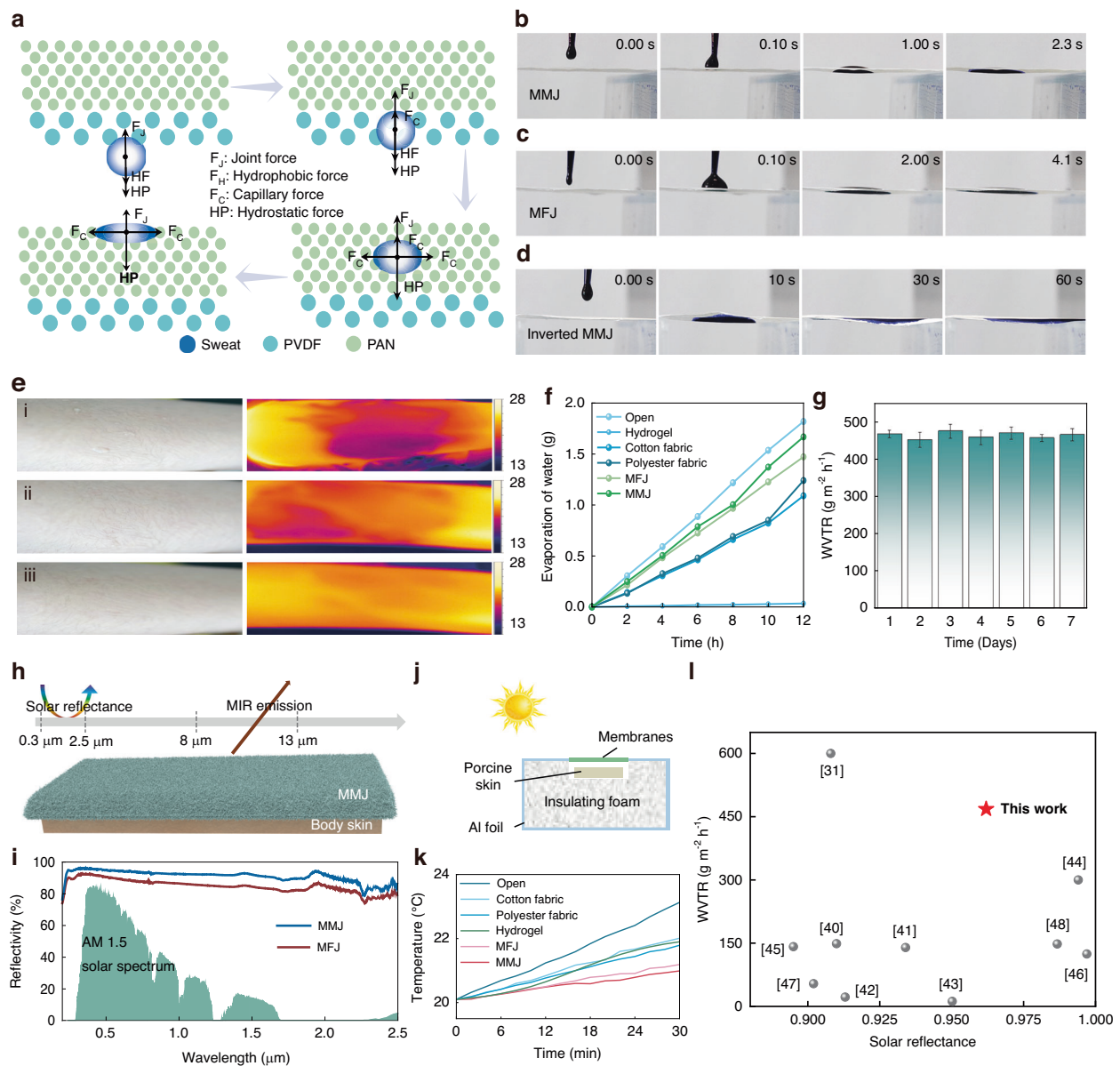


Fig. 3 Characterization of the wearable performance of the MMJ-based fabric. **a** Schematic diagram illustration of the principle of anti-gravity water transport. Directional water transport behavior on MMJ (**b**), MFJ (**c**), and inverted MMJ (**d**) membranes. **e** Photograph and infrared thermography images of the arm after 10-min running worn with polyester fabric (i), cotton (ii), and MMJ (iii) membranes. Scale bar: 1 cm. **f**, Water evaporation characterization of hydrogel, polyester fabric, cotton fabric, MFJ, and MMJ, while container opening as the control. **g** WTR variations for MMJ over a 7-day period. **h** Illustration of radiative cooling mechanism using MMJ-based fabric. **i** Reflectance spectra of MMJ and MFJ across the solar wavelength range. **j** Schematic diagram of the outdoor cooling performance testing setup with different materials as the seal membrane. **k** Temperature variation curves of the porcine skin beneath different seal membranes. **l** Comparison of solar reflectance and WTR between MMJ and other wearable devices reported in previous studies

to the skin. Driven by the synergistic effects of hydrostatic pressure and upward joint force mainly from sweat gland pressure and wetting gradient, the sweat undergoes anti-gravity transport and reaches the upper hydrophilic PAN layer. Then, propelled by capillary force and wetting gradient, the sweat exhibits rapid diffusion and upward transportation in the PAN layer. Ultimately, the sweat

gradually evaporated and dried out the electrospun membranes. This Janus hierarchical architecture facilitates directional transport of perspiration, thereby enabling rapid sweat excretion and effectively preventing undesirable accumulation on the skin surface. Moreover, the MMJ exhibits enhanced hydrophilic/phobic properties and intensified the hierarchical gradient compared with

microbead-free Janus (MFJ) system, leading to promoted sweat/moisture-wicking efficiency. Figure 3b–d illustrates the dynamic evolutions of the fluid behavior of water droplets on the MMJ, MFJ and inverted MMJ hybrid membrane. The results revealed that water droplet (dyed blue) immediately spread into the hydrophilic PAN side and permeated downward to reach the bottom when dropped from the hydrophobic PVDF side of MMJ membrane. The droplet subsequently accumulated at the MMJ's lower surface before detaching as discrete droplets. The MMJ membrane exhibited a remarkable droplet conduction capability, achieving transmembrane droplet transport (50 μL) in just 2.3 s, which represents a significant improvement compared to the MFJ membrane (4.1 s). In contrast, when the droplet deposition sequence was reversed (from PAN to PVDF), it rapidly spread across the hydrophilic PAN surface but remained entirely confined to this layer, showing no detectable infiltration into the underlying PVDF layer even after prolonged observation (Figs. S2, 3).

The practical sweat-wicking performance of the MMJ membrane was then investigated. The subject was asked to wear the MMJ membrane and engage in a 10-min running, with commercial polyester and cotton fabrics used as controls. It was found that localized sweat accumulation was predominantly observed beneath hydrophobic polyester fabrics, whereas cotton-covered skin exhibited more extensive sweat dispersion. This phenomenon may be attributed to the hydrophilic properties of cotton fabric, which facilitates rapid sweat absorption and subsequent lateral wicking across the textile-skin interface. In contrast, skin areas covered with MMJ membrane demonstrated negligible sweat accumulation, attributable to the superior sweat-wicking capacity that enables rapid directional sweat transport away from the skin surface (Fig. 3e). The permeability was further measured according to the ASTM E96 standard^{32,33}. The MMJ membrane was employed to seal water-filled bottles for measuring water vapor transmission rates (WVTRs) through periodic weight measurements. The results demonstrated that the evaporation rate through the MMJ membrane approximated that of an open container, significantly surpassing MFJ and commercial polyester fabrics, cotton fabric, and hydrogel patches (Fig. 3f). The calculated WVTR reached approximately 468.9 $\text{g}/\text{m}^2/\text{h}$, which is much higher than that of human skin at rest and exercise status^{34–36}. This merit indicates excellent breathability and sweat management capability, which is beneficial for prompt evaporation of residual moisture and humidity within the wearable system. Crucially, the MMJ membrane maintained consistent vapor transmission performance over extended durations (Fig. 3g), attributable to the stable regulation performance on the hydrophilic/hydrophobic properties of the electrospun

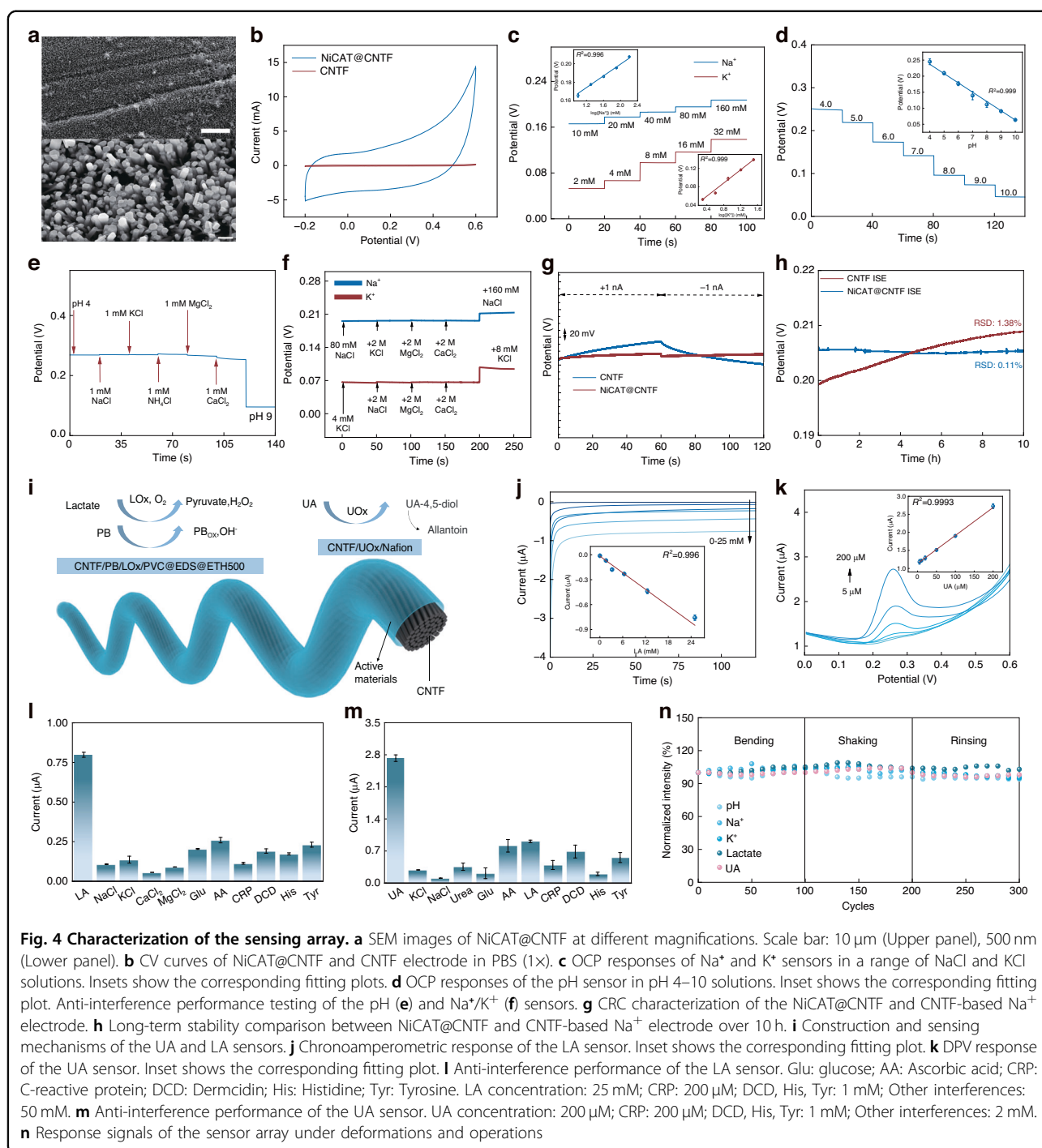
membrane by the microbead-mediated structural configuration, facilitating maintaining a dry and comfortable microclimate for on-body sensing application.

Achieving skin cooling is of great significance for outdoor activities, especially in intensive exercise settings. To realize effective radiative cooling, wearable devices are typically required to exhibit high reflectance in the solar spectrum (0.3–2.5 μm) and high emissivity within the mid infrared (MIR) atmospheric transparency window (8–13 μm) (Fig. 3h)^{37,38}. We systematically characterized the optical properties of MMJ and MFJ membranes. The results demonstrate that the MMJ membrane achieves a solar reflectance of 96.2% across the solar spectrum (Fig. 3i), coupled with an MIR atmospheric window emissivity of 91.3% (Fig. S4). These values represent a significant enhancement in radiative cooling performance compared to the MFJ membrane, which can be attributed to the additional dielectric resonance effects induced by the ellipsoidal bead structures within the MMJ membrane³⁹. Further practical radiative cooling performance of the MMJ was investigated using porcine skin. As shown in Fig. 3j, the temporary skin temperature was recorded while different membranes were utilized to cover the insulating porcine skin. The results found that the temperature of the porcine skin elevated by only 0.8 $^{\circ}\text{C}$ when it was covered by the MMJ membrane and exhibited a maximum temperature reduction of 2.1 $^{\circ}\text{C}$ compared with the uncovered counterpart, which surpasses that observed with alternative configurations utilizing the MFJ membrane, commercially available fabrics and hydrogel patches (Fig. 3k).

Overall, the MMJ membrane demonstrates excellent sweat-wicking capacity and superior radiative cooling performance, outperforming or at least matching wearable sensors developed in prior reported studies (Fig. 3l)^{31,40–48}. Furthermore, owing to the well-engineered porous architecture of the electrospun membrane and woven fabric layer, the MMJ membrane exhibits exceptional air permeability (13 mm/s). Consequently, this MMJ membrane achieves outstanding wearable functionality, providing unparalleled user comfort during exercise monitoring.

Characterization of the multiplex electrochemical biosensors

All sweat sensing systems consist of a working electrode and a reference electrode based on carbon nanotube fibers. Na^+ and K^+ quantification is achieved through ion-selective membranes, while pH measurement utilizes polyaniline surface protonation-induced potential changes⁴⁹. While these methods provide ion-specific responses, long-term operational stability remains challenging due to interfacial contact susceptibility at electrodes, particularly during extended exercise monitoring. In this regard, a layer of conductive metal-organic framework (MOF) material



based on Nickel triphenylene-fused metal catecholate (NiCAT) was modified on the Na^+ , K^+ , and pH working electrode. Its π -d conjugated structure coupled with inherent porosity confer not only a substantial specific surface area but also improve the double-layer capacitance, which is beneficial for promoting the sensitivity and mitigating potential drift for prolonged functionality⁵⁰. The CNTF substrate's inherent hydrophilicity and textured

surface morphology facilitated controlled crystallization of NiCAT, yielding well-aligned nanowire arrays. The SEM images (Fig. 4a, Fig. S5) demonstrated densely packed hexagonal prismatic nanowire structures. Elemental mapping (Fig. S6) confirms uniform distribution of C, O, and Ni across the CNTF surface, while HRTEM analysis (Fig. S7) verifies the crystalline nature of detached NiCAT nanowires. The X-ray photoelectron spectroscopy (XPS)

revealed high degree of coordination between the Ni ions and hydroxyl groups, and the characteristic peak in the high-resolution element spectrum was consistent with previous research (Fig. S8)^{51,52}. These results collectively confirming successful modification of the conductive MOF on the electrode. Electrochemical evaluation demonstrated significant performance enhancement in NiCAT@CNTF electrodes. Cyclic voltammetry (CV) in PBS solution showed expanded current response (Fig. 4b), with supplementary capacitance measurements (Fig. S9) revealing substantially increased double-layer capacitance compared to unmodified electrodes. These improvements directly contribute to enhanced sensor sensitivity and operational stability—key metrics for practical implementation in continuous sweat monitoring applications.

The analytical performance of each sensor was systematically evaluated through exposure to analyte solutions across physiological concentration ranges. Figure 4c and 4d shows the open-circuit potential (OCP) responses of the Na⁺, K⁺, and pH sensor in solutions containing 10–160 mM Na⁺, 2–32 mM K⁺, and pH 4–10, respectively. The results indicated that the ion-targeted sensors exhibited linear detection performance for Na⁺, K⁺, and pH, with sensitivities of 0.032 V/dec, 0.071 V/dec and 0.029 V/dec, respectively. The detection limits (LODs) are calculated to be 1.07 mM and 0.14 mM for Na⁺ and K⁺ sensors (*S/N* = 3), respectively. In order to further investigate the anti-interference capabilities of the three sensors, exogenous electrolytes were introduced into test solutions while monitoring signal variations (Fig. 4e, f). The experimental results showed that these three sensors have good anti-interference capabilities against common sweat matrix components. Moreover, the stability of NiCAT composite electrodes was rigorously examined through current-reversal chronopotentiometry (CRC) approach (Fig. 4g). As exemplified by Na⁺ ion selective electrode (ISE), the results confirmed that NiCAT modified ISE exhibited a stable potential response (7.1 nV/s) when polarized with a steady-state current for 60 s, while the unmodified electrode exhibited notable potential drift (1614 nV/s). Long-term stability assessments over 10 h (Fig. 4h, Fig. S10) further confirmed the superior performance of NiCAT@CNTF ISEs, showing minimal potential drift (RSD = 0.11%) compared to CNTF ISEs (RSD = 1.38%), highlighting the remarkable stability enhancement achieved through NiCAT modification.

The LA and UA concentration was determined based on enzyme-mediated reactions (Fig. 4i). To improve the anti-interference capability and operational stability, an external diffusion-limiting layer (PVC@DOS@ETH500) was introduced to encapsulate the LA sensor for regulating substrate permeation⁵³. Meanwhile, the UA sensor was modified with a layer of Nafion, facilitating the immobilization of enzymes and repelling negative-

charged interferents⁵⁴. Figure 4j, k illustrated the distinct linear responses of the LA and UA sensors toward LA from 0 to 25 mM, UA from 5 to 200 μM. The LA and UA sensors exhibited sensitivity of 33.27 nA/mM and 7.76 nA/μM, with an LOD of 0.52 mM and 1.68 μM (*S/N* = 3), respectively. Moreover, both sensors exhibited satisfactory selectivity against common sweat interferents including small-molecule metabolites and macromolecular substances (Fig. 4l, m), which is ascribed to the specific catalytic merit of enzymes. Additionally, artificial sweat spiked with different concentrations of biomarkers were further analyzed, revealing that the sweat sensors demonstrated brilliant detection recovery with relative standard deviation (RSD) values less than 10% (Table S1).

The signal response of the sensor array from different batches was further characterized. The results revealed that the sensor array maintained consistent signal response across batches, indicating excellent reproducibility (Fig. S11). Moreover, it was found that all the sensors showed consistent responses toward the same target during a 2-week storage period, which is beneficial for long term applications (Fig. S12).

In addition, we investigate the influence of environmental factors, including temperature and pH, on sensor performance to evaluate the robustness of the sensing system. As illustrated in Fig. S13, pH variations exerted a negligible effect on the output of the sensor array, primarily attributed to the protective layer modification of the electrode and the encapsulation of the fiber-based reference electrode. Temperature exhibited a marginal influence on the ion sensors. However, it had a relatively more pronounced effect on the LA and UA sensors, mainly due to the thermal sensitivity of enzymatic activity (Fig. S14). Nevertheless, under typical conditions, temperature fluctuations are generally modest and thus do not substantially compromise sensor functionality in the practical scenarios.

Additionally, we assessed the mechanical and operational durability under various deformation and post-treatment conditions. As shown in Fig. 4n, all five sensors remained consistent performance after 100 cycles of bending, shaking, and rinsing. Moreover, no significant variation in sensor output was observed during progressive stretching or bending (Figs. S15 and S16), underscoring the mechanical robustness of the fabric system. This stability is attributed to the low elastic modules of the CNTF, as well as the sparse integration density of the fabric network.

The comprehensive comparative analysis of our ion sensors and metabolite sensors against current research works has been performed as presented in Tables S2 and S3. It can be found that our ion sensors exhibited performance at least comparable with similar works, which is mainly attributed to the high specific surface area enabled by the unique

porous structure of the NiCAT. Moreover, benefiting from the great conductivity and 1D architecture of the CNTF, the LA and UA sensors could also exhibit satisfactory sensing performance while ensuring favorable flexibility and compactness.

Exercise healthcare using the fabric composite sensing system

For practical wearable applications, this MMJ-composite fabric can be connected to a flexible printed circuit board (FPCB) to realize wireless signal detection. Due to the excellent flexibility and biocompatibility, the fabric system is able to be worn on different positions on the human body such as the forehead, neck, and arm with extraordinary conformal contact interface (Fig. 5a). Moreover, an accessory commercial photoplethysmography (PPG) module was incorporated into the system to detect HR and SpO₂ by signal postprocessing, and an NTC thermal resistor was also embedded in the module for real time body temperature feedback. The signals can be detected on-demand and transmitted to a software interface in the mobile phone by the microcontroller unit and Bluetooth module (Fig. 5b), enabling multiplex biomarker monitoring in a real time and in situ manner. To evaluate the on-body sensing performance of our wearable system, we recruited five subjects and monitored sweat biomarker levels during exercise using the fabric sensing system. Concurrently, sweat samples were collected and determined by commercial detection devices such as enzyme-linked immunosorbent assay (ELISA) kits, to validate the outputs obtained from the wearable system. As shown in Fig. S17, the detection results from our wearable system exhibited no significant differences compared to those from the commercial devices ($p < 0.05$), demonstrating the reliable monitoring capabilities of our wearable system.

We next evaluated the dynamics of biochemical biomarkers during different exercise modalities using this fabric sensing system. The protocol included 10-min badminton (an aerobic-anaerobic hybrid activity), 5-min indoor cycling (moderate-high intensity aerobic exercise), and 15-min running (10-min jogging + 1-min sprint + 4-min rest). As shown in Fig. 5c, during badminton, a transient body temperature decrease occurred at ~2 min, likely attributable to evaporative cooling from initial perspiration. Electrochemical detection of sweat components commenced at 2.5 min, showing progressive increases in Na⁺, K⁺, LA, and UA concentrations, while sweat pH exhibited a descending trend potentially mediated by LA-induced acidification. HR surged from 71 to 170 bpm within the initial 4 min before stabilizing around 160 bpm. Most biomarkers demonstrated rapid initial changes followed by plateauing trends, mirroring cardiovascular adaptation patterns. During bicycle exercise, body

temperature showed immediate elevation (2 min) without subsequent cooling phase, possibly due to sustained heat production from continuous muscle activation. Na⁺ and K⁺ exhibited greater variations compared to badminton, with LA/UA maintaining upward trajectories and pH showing persistent decline. Both body temperature and HR reached elevated plateaus within 2 min post-exercise initiation. Jogging can be regarded as a typical low-intensity aerobic exercise. It displayed delayed thermal response, with modest temperature reduction emerging at 3.5 min—later onset than badminton, potentially reflecting lower metabolic demand. Na⁺ and UA showed mild increases while K⁺ exhibited paradoxical decrease, possibly indicating differential sweat gland reabsorption mechanisms. LA and pH remained stable throughout. HR reached a plateau at 134 bpm following an initial doubling (from 66 to 134 bpm within 2 min), with temperature eventually stabilizing slightly above initial level. Sprint intervals (anaerobic phase) induced marked biomarker fluctuations, particularly LA surging from 5.4 mM to 7.8 mM within 1 min, demonstrating rapid glycolytic metabolism. During the subsequent rest period, all the biomarkers gradually returned to baseline, and approached pre-exercise levels after 4 min with the exception of pH, LA and UA. Notably, SpO₂ showed a significant decline only post-sprint, remaining stable during other protocols. This multiplex biomarker monitoring revealed exercise-specific biomarker signatures and successfully captured their dynamic profiles, confirming the utility of the fabric composite sensing system for real-time exercise physiology monitoring and metabolic state analysis.

Further, the correlation between these biomarkers and exercise-induced fatigue was investigated. Continuous biomarker monitoring was conducted using the wearable fabric composite sensing system during diverse physical activities. Concurrently, the Borg Rating of Perceived Exertion (RPE) scale was implemented for quantitative fatigue assessment⁵⁵, establishing a three-tier classification outputs: low fatigue (RPE ≤ 2), medium fatigue (RPE 3–6), and high fatigue (RPE ≥ 7). Over a 7-day monitoring period, 168 paired datasets were systematically collected. A principal component analysis (PCA) revealed the clustering of the samples from the same group, indicating the features learned by the model are universal across individuals (Fig. S18). A support vector machine (SVM)-based machine learning classification model was subsequently developed to evaluate fatigue states through multimodal biosignal integration (Fig. 5d). The results demonstrated a robust correlation between biomarker profiles and fatigue gradations. The optimized classification model achieved an overall accuracy of 92.9%, with misclassifications limited to two high-fatigue and one low-fatigue cases erroneously categorized as medium fatigue state (Fig. 5e). Comparatively, the recognition model based on random

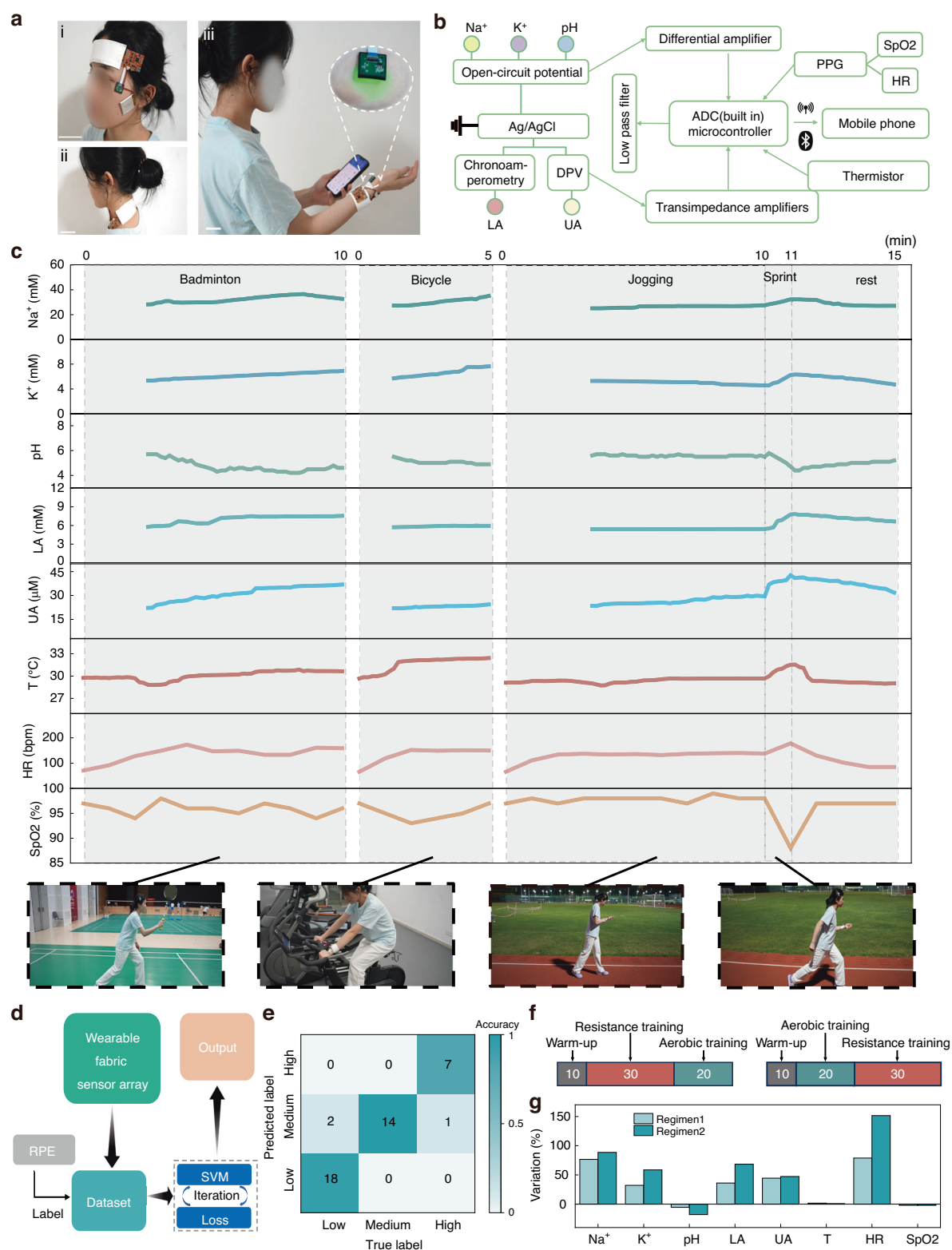


Fig. 5 Practical exercise analysis using the MMJ-based wearable fabric system. **a** Photographs of the integrated fabric system worn on a volunteer's forehead (i), neck (ii), and wrist (iii). Inset shows the PPG module for HR and SpO₂ detection. **b** Circuit framework of the wearable fabric system. **c** Multiplex biomarker monitoring during different exercise activities. Scale bar: 20 cm. **d** Schematic illustration of fatigue state assessment using the wearable fabric system and the machine learning model. **e** Confusion matrix of the fatigue assessment results. **f** Schematic illustration of two different exercise regimens. **g** Variation of different biomarker levels after exercise with different regimens

forest and XGBoost exhibited a relatively poor performance for fatigue state classification, with an overall prediction accuracy of 81.0% and 73.8%, respectively (Fig. S19). Moreover, a Leave-One-Out cross validation (LOOCV) strategy was utilized to evaluate the model generalization capability. The results show that the final accuracy was 94.6% with a standard deviation of 6.8%, and the model didn't completely fail on any specific fold, indicating a robust performance across different datasets. These findings demonstrated the feasibility of developing personalized fatigue assessment models through adaptive machine learning frameworks and multimodal biosensing technologies.

Moreover, we investigated the dynamic variations in biomarkers under different exercise protocols. Two distinct exercise regimens were designed (Fig. 5f): Regimen 1 consisted of a 10-min warm-up followed by 30 min of resistance training and 20 min of aerobic exercise, while Regimen 2 comprised a 10-min warm-up preceding 20 min of aerobic exercise and 30 min of resistance training. The wearable fabric sensing system was employed to monitor multiple biomarker levels pre- and post-exercise. Notably, all measured parameters except body temperature and SpO₂ demonstrated greater variation under regimen 2, with particularly significant alterations observed in LA concentration and HR levels (Fig. 5g, Fig. S20). This phenomenon could be mechanistically explained through metabolic pathway prioritization: Resistance training, as an anaerobic activity, induces substantial LA accumulation and rapid elevation of HR to elevated plateau levels when performed with adequate glycogen reserves. Subsequent aerobic exercise facilitates LA clearance while maintaining HR within fat-burning zones. Conversely, performing aerobic exercise prior to resistance training may deplete glycogen reserves during the initial phase, potentially leading to premature fatigue during subsequent resistance training. This glycogen depletion could exacerbate LA accumulation through compensatory anaerobic metabolism and drive HR beyond anaerobic thresholds due to cumulative cardiovascular stress. Supporting these physiological observations, the RPE values recorded 5 and 7 for regimen 1 and 2, respectively. The machine learning classification models further categorized the two post-protocol states as medium fatigue for regimen 1 and high fatigue for regimen 2, respectively, demonstrating strong concordance with the biomarker variation patterns. These findings collectively suggested that exercise regimen design significantly influences metabolic pathway activation and physiological stress responses, and wearable sensing system could provide reliable guidance for scientific regimen schedule.

Conclusion

In this study, we developed an advanced wearable fabric system by incorporating fiber sensor arrays with Janus

electrospinning membranes for exercise health management. The fiber sensor arrays enable sensitive detection of sweat biomarkers including Na⁺, K⁺, pH, LA, and UA with exceptional stability and selectivity. A commercial module based on PPG and thermistor was also integrated for simultaneous detection of physical indicators including body temperature, HR, and SpO₂, facilitating multimodal monitoring of exercise physiological dynamics. Beneficial from the merit of the multiplex sensing configuration and specific machine learning models, exercise health status can be precisely analyzed, facilitating scientific management and dynamic adjustment.

Moreover, the porous architecture of the woven textile and electrospun membranes confers excellent breathability of the fabric system. Microbead-mediated structure was optimized to regulate the interfacial hydrophilicity/hydrophobicity and optical characteristics of the electrospun membranes, which can facilitate outstanding wet permeability performance with a WVTR of 468.9 g/m²/h and radiative cooling performance with a solar reflectance of 96.2%. These optimized wearable performances facilitate the sensing system possessing great user comfort and utility in practical exercise activities, which is beneficial for improving user adaptability and extending the effective service of wearable sensors. In addition, this presents a promising application pathway for specialized exercise scenarios characterized by significant thermogenesis or perspiration. The wearable fabric system demonstrated reliable capability for multiplex biomarker monitoring in diverse exercise scenarios and discerning activity-specific physiological signatures. An identification accuracy of 92.9% was realized using the model, emphasizing the potential for personalized fatigue state monitoring using wearable biosensors. In addition, the fabric system demonstrated proficiency in exercise regimen evaluation, which establishes a technological foundation for data-driven optimization of training protocols. Our work presents a fabric-based wearable platform for exercise monitoring with excellent sensing functionality and wear comfort, offering a practical solution for effective and reliable exercise health management using wearable electronics and physiological profiling. In our future work, we will focus on integrating more accessible biochemical indicators and improving system integration, while also expanding our dataset to develop more accurate and robust machine learning models, ultimately enabling reliable health status assessment in real-world applications.

Methods

Materials and reagents

PAN, PVDF, N,N-Dimethylformamide (DMF), tetrahydrofuran (THF), 2,3,6,7,10,11-hexahydroxytriphenylene (HHTP), sodium ionophore X, potassium tetrakis[3,5-bis(trifluoromethyl)phenyl]borate(K-TFPB), tetrakis[3,5-

bis(trifluoromethyl)phenyl] boron sodium (Na-TFPB), bis(2-ethylehexyl) sebacate (DOS), polyvinyl chloride (PVC), polyvinyl butyral (PVB), Nafion, nickel(II) acetate tetrahydrate ($\text{Ni}(\text{OAc})_2 \cdot 4\text{H}_2\text{O}$), valinomycin, block polymer PEO-PPO-PEO (F127), methanol, NaCl, KCl, HCl, CaCl_2 , FeCl_3 , NH_4Cl , $\text{K}_3[\text{Fe}(\text{CN})_6]$, chitosan, glucose, urea, Ascorbic acid (AA), single-wall carbon nanotubes (SWCNTs), glutaraldehyde and Ag ink were purchased from Adamas Reagent (Shanghai, China). Phosphate Buffered Saline (1×) and bovine serum albumin (BSA) were purchased from Aladdin Reagent (Shanghai, China). CNT fibers were obtained from XFNANO (Nanjing, China). Syringes and spinnerets were purchased from Jinrong Electronic Chemical Technology Co., Ltd (Suzhou, China). Commercial polyester cotton was obtained from Diguan Clothing Accessories (Jiaxing, China). All chemical reagents are of analytical grade or above. Water was deionized before use (18.25 MΩ cm).

Preparation of the NiCAT@CNTF electrode

A mixture of 20 mg $\text{Ni}(\text{OAc})_2 \cdot 4\text{H}_2\text{O}$ and 14 mg HHTP was added to a 20 mL container and uniformly dispersed in 10 mL deionized water under ultrasonic treatment for 15 min. Subsequently, the CNTF was immediately immersed in the solution, the container was sealed, and the mixture was heated in an isothermal oven at 80 °C for 12 h. After cooling to room temperature, the NiCAT@CNTF electrodes were thoroughly washed with deionized water and ethanol multiple times and dried in an isothermal oven at 80 °C for 12 h to complete the fabrication process.

Preparation of the sensing electrode array

All five biomarkers were tested using a “two-electrode” system consisting of a working electrode and reference electrode.

Preparation of the fiber reference electrode

Silver ink was drop-cast onto the CNT fibers, which were then treated with a 0.1 M FeCl_3 solution for 1 min. The PVB mixture was prepared by dissolving 79.1 mg of polyvinyl butyral (PVB), 50 mg of sodium chloride (NaCl), 2.0 mg of F127, and 0.1 mg of CNT in 1 mL of methanol. Subsequently, a drop of a PVB mixture was coated on the RE to reduce the potential drift.

Preparation of Na^+ working electrode

A mixture solution was prepared by dissolving 3 mg Na^+ ionophore X, 187.5 mg DOS, 108 mg PVC, and 1.5 mg Na-TFPB into 3 mL of THF. Then the solution (10 μL) was drop-casted on the NiCAT@CNTF and then dried at room temperature for 12 h.

Preparation of the K^+ working electrode

A mixture was prepared by dissolving 3 mg Valinomycin, 187.5 mg DOS, 108 mg PVC, and 1.5 mg K-TFPB in 3 mL THF. Then the solution (10 μL) was drop-casted on the NiCAT@CNTF and then dried at room temperature for 12 h.

Preparation of the pH working electrode

A layer of PANI was electrodeposited on the NiCAT@CNTF, which was achieved by scanning CV in 0.1 M aniline and 1 M HCl from −0.2 V to 1.0 V for 25 cycles at 100 mV/s.

Preparation of the LA working electrode

A layer of Prussian blue was electrodeposited onto the CNTF. The electrodeposition solution consisted of 2.5 mM ferric chloride, 2.5 mM potassium ferricyanide, 100 mM potassium chloride, and 100 mM hydrochloric acid. The deposition was performed within a voltage range of −0.2 V to 1.0 V at a scan rate of 100 mV s^{−1} for 50 cycles. Next, a mixture of 2% acetic acid solution, 1% chitosan solution, and 5 mg of SWCNTs were prepared and dispersed using ultrasonic treatment for 30 min. Subsequently, 4 mg of lactate oxidase (LOx) was added to the mixture. Finally, 4 μL of the prepared chitosan/SWCNT/ LOx mixture was evenly drop-coated onto the Prussian blue-coated CNTF. An external diffusion-limiting membrane (containing 33 mg/mL PVC, 66 mg/mL DOS, and 3 mg/mL ETH500) was drop-casted on the fiber electrode and dried at room temperature for 20 min.

Preparation of the UA working electrode

A solution of uricase (2 wt%) was mixed with a 4 wt% bovine serum albumin (BSA) solution and a 2 wt% glutaraldehyde solution in a volume ratio of 2:1:1 (V/V) to prepare the uricase/BSA/glutaraldehyde solution. Four microliters (4 μL) of the mixture were drop-coated onto the CNTF and dried overnight at 25 °C to obtain the uric acid sensor. To enhance stability, a 2 μL Nafion solution (0.5 wt%) was drop-cast onto the working electrode.

Preparation of the MMJ membrane

MMJ was prepared by electrospinning self-assembly technique. A 15 wt% PVDF solution was prepared by dissolving PVDF in a mixture of acetone and DMF (2:1, V/V) and stirring at 90 °C for 3 h. The PVDF solution was then electrospun at 1 mL/h flow rate, 10 kV applied voltage and 10 cm TCD. The fabric interwoven with fiber electrodes was immobilized on the collector. Subsequently, a 10 wt% PAN solution was prepared by dissolving PAN in DMF and stirring at 60 °C for 4 h. Using the PVDF deposited fabric as the collector, the PAN solution was electrospun at 1 mL/h flow rate, 17 kV applied voltage and 10 cm TCD. For microbead modulation, the

concentration and applied voltage in PVDF electrospun process were modulated from 8 wt% to 24 wt%, and 10 to 20 kV, respectively. The concentration and applied voltage in PAN electrospun process were modulated from 6 wt% to 16 wt%, and 17 to 25 kV, respectively. The environment temperature and humidity were set as 25 °C and 50% relative humidity. For the preparation of MFJ, all the craft parameters were similar to that of MMJ, except that the concentration of PVDF electrospinning solution is 24 wt% and that of PAN electrospinning solution is 16 wt%.

Integration of the fabric system

The hierarchical Janus membrane was obtained through layer-by-layer electrodeposition. The porous woven layer was embedded in the intermediate layer. All functional fibers, except for the electrode modification areas, were encapsulated with a polymer waterproof coating, and were connected at the ends to a microcontroller unit by pins for on demand signal detection, conversion, and transmission. A self-assembled commercial PPG module was connected with the FPCB while using a green light (530 nm) for HR measurement, a red light (660 nm) and an infrared (940 nm) for SpO₂ detection. Notably. The red light can also be utilized for rough HR determination. A thermistor (NTC 0402) was also integrated to monitor the temperature in a high resolution. A Bluetooth module was installed for wireless result transmission between the on-body sensor system and a customized App, which enables storage, analysis, and retrospect of detected data.

Characterization of materials and sensors

The WCA was measured using an optical contact angle meter (BDOU, CA 200). The microstructures of the materials were characterized by SEM (Quantum, 5000X +UltimMax 40e) and TEM (JEOL, F200). Tensile testing was performed using a nano-mechanical tester (Hengyi, HY 0580). All electrochemical measurements were conducted using a CHI 660e electrochemical workstation. The bead density was calculated with the aid of ImageJ 1.53t. Statistical analysis was performed using SPSS 22.0. Photographs were taken using a high-resolution camera (Sony, Alpha 6400). The air permeability values (mm s⁻¹) of all samples were tested using an air permeability tester (Runhu, RH-TQZ500). The thermal images and surface temperature distribution of the samples were recorded using an infrared thermal camera (Optris, PI 450i). The solar reflectance was measured using a UV-VIS-NIR spectrophotometer coupled with a diffuse integrating sphere (Shimadzu, UV-3600 plus). The MIR emissivity was measured using a Fourier-transform infrared spectrometer (Wanyi, IR2 dual-band emissivity measurement instrument) equipped with a diffuse gold integrating sphere. The Na⁺, K⁺, and pH sensing results were

validated by commercial Na⁺ analyzer (DWS-51, Leici), K⁺ analyzer (PKS-K, Qiwei), and pH meter (PHSJ-3F, Leici). The LA and UA sweat samples were in vitro determined by ELISA kits (Adamas Life).

WVTR measurement

The experimental procedure was conducted according to the ASTM E96 standard: Deionized water was placed into a 10 mL bottle featuring a circular opening approximately 2 cm in diameter. The bottle was sequentially sealed using four distinct materials: MMJ, MFJ, cotton fabric, and polyester fabric, respectively. The complete test assembly was then positioned in a controlled environment maintained at 40 °C for a duration of 12 h. The mass of each sealed component was measured at 1 h intervals throughout the experiment. The WVTR was subsequently determined by calculating the ratio of mass loss to the exposed open area of the sealing materials.

In vitro measurement of radiative cooling performance

Porcine skin was employed as a simulated skin substrate, with a layer of thermal insulation foam positioned beneath it to minimize heat dissipation towards the base. The entire apparatus was subsequently encased in a layer of aluminum foil to reflect incident solar radiation. Four distinct material samples—MMJ, MFJ, cotton fabric, and polyester fabric—were individually placed over the simulated skin surface. Surface temperature measurements of the skin analog were systematically acquired at 2 min intervals using a calibrated infrared thermometer.

Human subject recruitment

All study participants were recruited from the Harbin Institute of Technology, Shenzhen campus. Individuals with specific health conditions, including fractures, cardiovascular diseases, hypertension, or other medically relevant comorbidities, were excluded from this investigation. Informed consents were obtained from all study subjects before enrollment in the study. The experimental protocol strictly adhered to the ethical principles outlined in the World Medical Association's Declaration of Helsinki and received formal approval from the Research Ethics Committee of Huazhong University of Science and Technology Union Shenzhen Hospital.

Exercise monitoring with different scenarios

Volunteers were asked to sequentially perform four exercise trials with the equipment of the composite wearable fabric system: 10-min badminton, 5-min bicycle, 10-min jogging, and 1-min high-intensity sprint. The sensor system employed a dual-frequency

data acquisition protocol: physiological parameters including body temperature and sweat biomarker concentrations were recorded at 10-s intervals, while SpO₂ and HR measurements were synchronously captured every minute throughout the exercise sessions. Subjects were provided with adequate recovery time (exceeding 3 h) following badminton and bicycle activities to ensure their physiological state returned to baseline levels prior to exercise.

Machine learning-assisted fatigue assessment

The volunteer was instructed to engage in physical exercise for a total of 4 h each day at their convenience. After every 10-min exercise session, biomarker levels were collected using composite fabric system, and the RPE scores were recorded simultaneously. This resulted in a total of 168 data sets after one week. The data were labeled based on RPE: values between 0 and 2 were classified as low fatigue, 3–6 as medium fatigue, and 7–10 as high fatigue. The dataset was divided into training and testing sets in a 3:1 ratio. A classification model was constructed using the SVM algorithm, with all data normalized prior to model input. LOOCV strategy was employed to characterize the model generalization capability. The data collected from same period were utilized as the test dataset and the remaining data served as the training dataset. Each data will be selected once as the test dataset. The performance metrics including final accuracy and standard deviation were averaged across all folds. The algorithm was compiled using Python 3.9.

Exercise regimen evaluation

Under strictly controlled experimental conditions with all other variables maintained constant, volunteers performed two exercise protocols in sequence: (1) 10-min warm-up followed by 30 min of resistance training and 20 min of aerobic exercise; (2) 10-min warm-up preceding 20 min of aerobic exercise and 30 min of resistance training. The resistance training consists of a circuit comprising plank holds, dumbbell lifts, and weighted squats, with each circuit lasting 10 min. A total of three circuits are performed, with a 2-min rest interval between each circuit. The aerobic exercise was conducted with treadmill running. The composite fabric system was utilized to detect the biomarker signals before and after the exercise period. RPE scores were simultaneously recorded.

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Data availability

The data that support the findings of this study are available from the corresponding author on reasonable request.

Conflict of interest

The authors declare no competing interests.

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References

- Nyberg, S. T. et al. Health benefits of leisure-time physical activity by socio-economic status, lifestyle risk, and mental health: a multicohort study. *Lancet Public Health* **10**, e124–e135 (2025).
- Posadzki, P. et al. Exercise/physical activity and health outcomes: An overview of Cochrane systematic reviews. *BMC Public Health* **20**, 1724 (2020).
- Das, P. & Horton, R. Physical activity-time to take it seriously and regularly. *Lancet* **388**, 1254–1255 (2016).
- Ates, H. C. et al. End-to-end design of wearable sensors. *Nat. Rev. Mater.* **7**, 887–907 (2022).
- Sempionatto, J. R., Lasalde-Ramírez, J. A., Mahato, K., Wang, J. & Gao, W. Wearable chemical sensors for biomarker discovery in the omics era. *Nat. Rev. Chem.* **6**, 899–915 (2022).
- Chen, S. et al. Flexible wearable sensors for cardiovascular health monitoring. *Adv. Healthc. Mater.* **10**, 2100116 (2021).
- Derbyshire, P. J., Barr, H., Davis, F. & Higson, S. P. J. Lactate in human sweat: a critical review of research to the present day. *Physiol. Sci.* **62**, 429–440 (2012).
- Gao, F. et al. Wearable and flexible electrochemical sensors for sweat analysis: a review. *Microsyst. Nanoeng.* **9**, 1 (2023).
- Davis, N., Heikenfeld, J., Milla, C. & Javey, A. The challenges and promise of sweat sensing. *Nat. Biotechnol.* **42**, 860–871 (2024).
- Shinde, S. et al. Wearable sweat-sensing patches for non-invasive and continuous health tracking. *Sens. Actuators Rep.* **9**, 100265 (2025).
- Lin, B. et al. Monitoring of sweat ions and physiological parameters via a reconfigurable modular system. *ACS Sens* **9**, 1272–1279 (2024).
- Jiang, D. et al. Skin-interfaced wearable sensor for long-term reliable monitoring of uric acid and pH in sweat. *Nano Lett.* **25**, 1427–1435 (2025).
- Gherghina, M.-E. et al. Uric acid and oxidative stress—relationship with cardiovascular, metabolic, and renal impairment. *Int. J. Mol. Sci.* **23**, 3188 (2022).
- Gong, S., Lu, Y., Yin, J., Levin, A. & Cheng, W. Materials-driven soft wearable bioelectronics for connected healthcare. *Chem. Rev.* **124**, 455–553 (2024).
- Luo, Y. et al. Technology roadmap for flexible sensors. *ACS Nano* **17**, 5211–5295 (2023).

16. Wang, P. et al. The evolution of flexible electronics: from nature, beyond nature, and to nature. *Adv. Sci.* **7**, 2001116 (2020).
17. Chen, S. et al. Wearable flexible microfluidic sensing technologies. *Nat. Rev. Bioeng.* **1**, 950–971 (2023).
18. Gao, W. et al. Fully integrated wearable sensor arrays for multiplexed in situ perspiration analysis. *Nature* **529**, 509–514 (2016).
19. Chen, Y. et al. Large-scale flexible fabric biosensor for long-term monitoring of sweat lactate. *Adv. Funct. Mater.* **34**, 2401270 (2024).
20. Cai, X. et al. Fully integrated multiplexed wristwatch for real-time monitoring of electrolyte ions in sweat. *ACS Nano* **18**, 12808–12819 (2024).
21. Wang, J. et al. Design of self-healing nanocomposite hydrogels and the application to the detection of human exercise and ascorbic acid in sweat. *Biosens. Bioelectron.* **267**, 116767 (2025).
22. Sun, Y. et al. Stretchable and smart wettable sensing patch with guided liquid flow for multiplexed in situ perspiration analysis. *ACS Nano* **18**, 2335–2345 (2024).
23. Xue, J., Wu, T., Dai, Y. & Xia, Y. Electrospinning and electrospun nanofibers: methods, materials, and applications. *Chem. Rev.* **119**, 5298–5415 (2019).
24. Ji, D. et al. Electrospinning of nanofibers. *Nat. Rev. Methods Prim.* **4**, 1 (2024).
25. Fong, H., Chun, I. & Reneker, D. H. Beaded nanofibers formed during electrospinning. *Polymer* **40**, 4585–4592 (1999).
26. Zhao, H., Chi, H. Electrospun bead-on-string fibers: useless or something of value? in *Novel Aspects of Nanofibers* (Lin, T. ed) (Rijeka: IntechOpen, 2018).
27. Zuo, W. et al. Experimental study on relationship between jet instability and formation of beaded fibers during electrospinning. *Polym. Eng. Sci.* **45**, 704–709 (2005).
28. Shi, S. et al. Lotus leaf-inspired breathable membrane with structured microbeads and nanofibers. *ACS Appl. Mater. Interfaces* **14**, 39610–39621 (2022).
29. Jiang, L., Zhao, Y. & Zhai, J. A lotus-leaf-like superhydrophobic surface: a porous microsphere/nanofiber composite film prepared by electrohydrodynamics. *Angew. Chem. Int. Ed.* **43**, 4338–4341 (2004).
30. Zhang, Y. et al. Thermal and moisture managing e-textiles enabled by Janus hierarchical gradient honeycombs. *Adv. Mater.* **36**, 2311633 (2024).
31. Hu, M. et al. Janus membrane with oriented fiber structure for boosted personal wet-thermal management. *Chem. Eng. J.* **502**, 157995 (2024).
32. Cheng, Y., Wang, J., Lu, X. & Wang, C. An all-nanofibrous Janus textile with directional perspiration for triboelectric nanogenerator and self-powered e-skin sensor. *Nano Energy* **117**, 108852 (2023).
33. Kabir, S. B., Bouazza, A. & Faizal, M. Modifying ASTM E96 to assess water vapour transmission rates of geomembranes at high temperatures. *Geotext. Geomembr.* **52**, 1054–1058 (2024).
34. Yang, X. et al. All-nanofiber-based Janus epidermal electrode with directional sweat permeability for artifact-free biopotential monitoring. *Small* **18**, 2106477 (2022).
35. Taylor, N. A. S. & Machado-Moreira, C. A. Regional variations in transepidermal water loss, eccrine sweat gland density, sweat secretion rates and electrolyte composition in resting and exercising humans. *Extrem. Physiol. Med.* **2**, 4 (2013).
36. Machado-Moreira, C. A., Wilmsink, F., Meijer, A., Mekjavic, I. B. & Taylor, N. A. S. Local differences in sweat secretion from the head during rest and exercise in the heat. *Eur. J. Appl. Physiol.* **104**, 257–264 (2008).
37. Liu, J.-Z. et al. Large-area radiation-modulated thermoelectric fabrics for high-performance thermal management and electricity generation. *Sci. Adv.* **11**, eadr2158 (2025).
38. Han, T. et al. Advances in radiative sky cooling based on the promising electrospinning. *Renew. Sust. Energ. Rev.* **200**, 114533 (2024).
39. Kim, H., McSherry, S., Brown, B. & Lenert, A. Selectively enhancing solar scattering for direct radiative cooling through control of polymer nanofiber morphology. *ACS Appl. Mater. Interfaces* **12**, 43553–43559 (2020).
40. Miao, D., Cheng, N., Wang, X., Yu, J. & Ding, B. Integration of Janus wettability and heat conduction in hierarchically designed textiles for all-day personal radiative cooling. *Nano Lett.* **22**, 680–687 (2022).
41. Miao, D., Cheng, N., Wang, X., Yu, J. & Ding, B. Sandwich-structured textiles with hierarchically nanofibrous network and janus wettability for outdoor personal thermal and moisture management. *Chem. Eng. J.* **450**, 138012 (2022).
42. Gu, B., Fan, F., Xu, Q., Shou, D. & Zhao, D. A nano-structured bilayer asymmetric wettability textile for efficient personal thermal and moisture management in high-temperature environments. *Chem. Eng. J.* **461**, 141919 (2023).
43. Feng, M. et al. Integrated passive cooling fabrics with bioinspired perspiration-wicking for outdoor personal thermal management. *Compos. B Eng.* **264**, 110875 (2023).
44. Yan, Z., Zhu, G., Fan, D. & Li, Q. Bioinspired metafabric with dual-gradient Janus design for personal radiative and evaporative cooling. *Adv. Funct. Mater.* **35**, 2412261 (2025).
45. Hazarika, A. et al. Hierarchically designed 3-d printed porous nylon fabric-based personal thermoregulatory for radiative and directional wick-evaporative cooling. *Chem. Eng. J.* **471**, 144536 (2023).
46. Du, P. et al. A high-performance passive radiative cooling metafabric with Janus wettability and thermal conduction. *Small* **20**, 2403751 (2024).
47. Zhong, S. et al. Radiative cooling cellulose-based fabric with hierarchical structure for outdoor personal thermal management. *Chem. Eng. J.* **489**, 151482 (2024).
48. Zhao, Q. et al. Sustainable, biodegradable and flexible laminated passive cooling membrane for personal thermal and moisture comfort under direct sunlight. *Chem. Eng. J.* **507**, 160698 (2025).
49. Wang, L. et al. Weaving sensing fibers into electrochemical fabric for real-time health monitoring. *Adv. Funct. Mater.* **28**, 1804456 (2018).
50. Wang, S. et al. Vertically aligned conductive metal-organic framework nanowires array composite fiber as efficient solid-contact for wearable potentiometric sweat sensing. *Sens. Actuat. B-Chem.* **369**, 132290 (2022).
51. Shi, Y.-X. et al. Soft electrochemical actuators with a two-dimensional conductive metal-organic framework nanowire array. *J. Am. Chem. Soc.* **143**, 4017–4023 (2021).
52. Wu, H. et al. Conductive metal-organic frameworks selectively grown on laser-scribed graphene for electrochemical microsupercapacitors. *Adv. Energy Mater.* **9**, 1900482 (2019).
53. Xuan, X. et al. Fully integrated wearable device for continuous sweat lactate monitoring in sports. *ACS Sens.* **8**, 2401–2409 (2023).
54. Xu, Z. et al. Non-enzymatic electrochemical detection of uric acid with electrodeposited Nafion film. *J. Electroanal. Chem.* **841**, 129–134 (2019).
55. De Almeida, E. et al. Applying ubiquitous sensing to estimate perceived exertion based on cardiorespiratory features. *Sports Eng.* **24**, 9 (2021).